Huntington's Disease as Attractor Geometry

A Principles-First Derivation of the
Stage-Stratified Intervention Protocol
from the Identity Axis Framework:

The Trigger, the Hub, the Identity Anchor,
and the Geometric Case That an "Incurable" Disease
Is Calculably Curable at Every Stage

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Huntington's Disease is not incurable. It is a terminal false attractor in the Waddington epigenetic landscape of medium spiny neurons, driven by a known single-gene trigger, maintained by a known epigenetic Hub, operating on a known Identity Anchor, with a stage-stratified intervention protocol that is *geometrically complete* — and every tool required at every stage either already exists or is in active clinical development as of 2026-03-09. The disease is calculable. The intervention is derivable. The cure is achievable, defined precisely and stratified by stage.

Abstract

We apply the Identity Axis Framework — a principles-first, axiomatic model of disease as attractor failure in Waddington epigenetic state space — to Huntington's Disease (HD). We derive the cell of origin (medium spiny neurons, MSNs), the Identity Anchor (BCL11B/CBP/DARPP-32/BDNF-TrkB), and the Convergence Hub (HDAC2/3 acting unopposed at MSN identity loci due to CBP sequestration by mutant huntingtin, mHTT). We distinguish the *trigger* (mHTT) from the *Hub* (HDAC2/3 dominance) and show this distinction requires a two-target intervention strategy that the current clinical landscape has not yet assembled.

We identify three coexisting MSN populations at any disease stage (protected, rescuable, terminal) and derive a stage-stratified four-level intervention protocol: Level 0 (mHTT substrate removal), Level 1 (Hub inhibition via HDAC2/3 inhibitors), Level 1A (Identity Anchor restoration via BDNF-TrkB agonism and CBP rescue), and Level 2A (cell replacement via iPSC-derived MSN transplantation).

We show that: (i) for pre-manifest carriers, the protocol predicts *prevention of disease onset* using existing tools; (ii) for early-manifest patients, near-halt of progression is achievable with the Level 0 + Level 1 combination; (iii) for late-manifest

patients, a defined engineering pathway to cure exists. The current clinical ceiling — 75% slowing from AMT-130 alone (uniQure Phase 1/2, September 2025) — is the Level 0 ceiling; the geometry predicts it rises to ≥90% slowing with the full combination protocol.

Six falsifiable predictions are stated for the record, all timestamped 2026-03-09. Zero structural contradictions with the Identity Axis Theorem are present.

Keywords: Huntington's disease; Waddington landscape; attractor geometry; medium spiny neuron; CBP; HDAC; Identity Axis; epigenetic Hub; mHTT; BCL11B; DARPP-32; BDNF; cell therapy; neurodegeneration; principles-first derivation; pre-manifest prevention

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1. Preamble: Why This Disease Is Structurally Special

Every other disease in the Identity Axis Framework required the derivation to *find* the Identity Anchor and Hub from the geometry of the epigenetic landscape — to identify, without prior knowledge, which gene most constitutes the cell's identity and which gene most maintains the false attractor. Huntington's Disease hands you the answer.

It is the only major disease in the table where:

(a) The causal gene is **fully known** — a single gene, *HTT*, with a measurable CAG repeat expansion that predicts age of onset with quantitative precision;

(b) **Pre-symptomatic detection** is possible decades before onset via genetic testing, with 100% certainty;

(c) The molecular cascade is among the most precisely characterised in all of neurodegeneration;

(d) The cell of origin is **unambiguous** — medium spiny neurons (MSNs) of the caudate and putamen;

(e) The disease follows a **predictable trajectory** from a known starting condition: CAG repeat length predicts age of onset.

This is the framework's most constrained case. Most constrained means most tractable. The geometry is not obscured by heterogeneity, unknown origin cells, or ambiguous Hubs. It is maximally clear.

The central claim of this document: HD is geometrically curable — in the same sense that drag force is calculable: given the correct inputs (stage, geometry, intervention level), the outcome is derivable before any clinical data is examined. The tools for every level either exist or are in late-stage clinical development. The combination has not been assembled. Assembling it is not scientifically impossible. It is logistically and organisationally incomplete.

2. Framework Foundations

2.1. The Identity Axis Theorem

The Identity Axis Framework rests on five axioms [1].

Axiom 1 (Waddington Principle)**.** *Every normal cell type occupies a stable attractor in an epigenetic landscape. The attractor is defined by the gene regulatory network that maintains the cell's identity.*

Axiom 2 (False Attractor)**.** *Disease is a cell displaced from its normal attractor and stabilised in a false attractor, maintained by a convergent epigenetic mechanism.*

Axiom 3 (Identity Anchor)**.** *Every cell of origin has a defining identity programme $\mathcal{A}$ — the element whose activity most directly constitutes what the cell is.*

Axiom 4 (Convergence Hub)**.** *Every false attractor has a maintaining element $\mathcal{H}$ — whose activity most directly stabilises the displaced state.*

Axiom 5 (Identity Axis)**.** *The ratio*

$$R = \frac{\mathcal{A}}{\mathcal{H}} \tag{1}$$

is a continuous, measurable coordinate in the Waddington landscape. R predicts attractor depth. Attractor depth predicts clinical outcome. $\mathcal{H}$ is the primary drug target.

2.2. HD as a Special Case: Terminal False Attractor

Definition 1 (Terminal False Attractor)**.** *A false attractor is* terminal *if the cell displaced into it cannot survive indefinitely and progresses irreversibly toward apoptosis.*

Definition 2 (Stable False Attractor)**.** *A false attractor is* stable *if the displaced cell can survive and self-maintain indefinitely in the false state.*

In cancer, the false attractor is stable — the cancer cell lives, the Hub maintains it, and the cell is recoverable by Hub dissolution. In HD, the false attractor is **terminal**: the MSN cannot sustain the displaced state, attempts cell-cycle re-entry, and undergoes apoptosis. Every dead MSN is a permanent loss.

Corollary 1 (Intervention Window)**.** *The fraction of rescuable MSNs decreases monotonically with disease progression. The pre-manifest stage maximises the rescuable fraction. The intervention window is finite.*

3. The Geometric Mapping

3.1. Cell of Origin

MSNs comprise ≈95% of striatal neurons. Within this population, *indirect-pathway* D2-type MSNs (expressing DRD2 and PENK/enkephalin, projecting to external globus pallidus) degenerate first.

This ordering is geometrically predicted: D2 MSNs have lower baseline BDNF-TrkB signalling density than D1 MSNs [5], placing them closer to the identity maintenance threshold at baseline. Cells with the thinnest margin fall first. This is attractor depth ordering: the shallowest attractor yields first under Hub pressure.

3.2. The Identity Anchor ($\mathcal{A}$)

The MSN Identity Anchor operates at three nested levels.

3.2.1. Deep Identity Anchor: BCL11B / CTIP2

BCL11B is the master transcription factor for MSN identity specification. It controls differentiation of MSN progenitors from the lateral ganglionic eminence, governs all downstream MSN-selective gene expression, and represses alternative neuronal fates [9]. BCL11B loss in experimental models recapitulates core HD features: DARPP-32 loss, MSN identity marker loss, and striatal connectivity disruption [10, 11].

3.2.2. Operational Identity Anchor: CBP-Mediated Chromatin Accessibility

CREB-binding protein (CBP) is a histone acetyltransferase (HAT) that maintains chromatin accessibility at BCL11B, DARPP-32, BDNF, and PGC-1α loci via H3K27 and H3K9 acetylation. Without CBP HAT activity at these loci the MSN Identity Anchor programme cannot be maintained regardless of which transcription factors are present.

3.2.3. Trophic Identity Anchor: BDNF / TrkB

BDNF is produced in the cortex and transported anterogradely to the striatum, where it activates TrkB receptors on MSNs. Loss of TrkB signalling in indirect-pathway SPNs triggers GSTO2 upregulation, leading to aberrant glutathione metabolism, elevated striatal dopamine, and dopaminergic neurotoxicity to MSNs [5]. This cascade precedes clinical onset.

The Identity Anchor in ratio terms is operationally measured as:

$$\mathcal{A} = \underbrace{\text{H3K27ac} + \text{H3K9ac}}_{\substack{\text{at BCL11B, DARPP-32,} \\ \text{BDNF, PGC-1}\alpha\text{ loci}}} \tag{2}$$

Clinical surrogate: DARPP-32 expression and CSF BDNF levels.

3.3. The Convergence Hub ($\mathcal{H}$)

Critical distinction — mHTT is not the Hub.
mHTT is the *trigger*: the upstream force that initiates the cascade. The Hub is the element that *maintains* the false attractor once displacement has begun. These are different molecular entities. Targeting only the trigger is necessary but not sufficient.

The Hub is: **HDAC2/3 acting unopposed at MSN identity loci due to CBP sequestration by mHTT.**

The molecular cascade:

$$\underbrace{\text{mHTT (polyQ)}}_{\text{trigger}} \xrightarrow{\text{binds CBP}} \underbrace{\text{CBP sequestered}}_{\text{in aggregates}} \xrightarrow{\text{HAT absent}} \underbrace{\text{HDAC2/3 unopposed}}_{\text{at MSN loci}}$$

$$\longrightarrow \underbrace{\text{H3K27ac}\downarrow\ /\ \text{H3K9ac}\downarrow}_{\text{BCL11B, DARPP-32, BDNF, PGC-1}\alpha} \xrightarrow{R\to 0} \underbrace{\text{cell-cycle re-entry}}_{\text{terminal false attractor}} \longrightarrow \text{apoptosis}$$

Literature confirmations:

- HDAC2/3 selective knockout in MSNs partially corrects HD transcriptional pathology [6];
- Vorinostat (SAHA), brain-penetrant pan-HDAC inhibitor, extends survival and improves motor behaviour in HD mouse models [7];

- HDAC6-selective inhibitor CKD-504 reduces mHTT aggregation *and* restores synaptic function in YAC128 HD mice [8];
- Selective HDAC3 inhibition rescues synaptic and transcriptional phenotypes in HD models [16].

4. The Three MSN Populations

At any disease stage, three coexisting MSN populations are derivable from the geometry. Their proportions shift monotonically with progression (Table 1).

4.1. Population 1 — Protected (R above threshold)

R is above the identity maintenance threshold. DARPP-32 is reduced but not lost. The cell is stressed but alive and identified.

These cells can be fully protected. Remove the trigger; the cascade does not advance; R stabilises; the cell survives with identity intact.

4.2. Population 2 — Rescuable (R below threshold, cell alive)

R has fallen below threshold. HDAC2/3 dominance is active. DARPP-32 and BCL11B are significantly reduced. The cell approaches the terminal threshold but has not crossed it. Chromatin at MSN identity loci is closed but not irreversibly so.

These cells can be rescued. If Hub dominance is removed AND the Identity Anchor signal is restored, R rises back above threshold. This is the mechanistic basis for the 75% slowing with AMT-130 alone [3]: Population 2 cells partially rescued by trigger removal even without explicit Hub inhibition. Level 1 Hub inhibition should rescue significantly more.

4.3. Population 3 — Terminal

R has collapsed. Cell-cycle re-entry has occurred. Apoptosis is underway or complete.

These cells cannot be rescued by Levels 0–1A. The intervention is replacement.

Table 1: MSN population distribution by disease stage (geometric estimate)

Stage	Pop. 1 (%)	Pop. 2 (%)	Pop. 3 (%)
Pre-manifest	95–99	1–5	≈0
Early manifest	60–75	20–35	5–10
Mid-manifest	20–40	20–30	30–50
Late manifest	5–15	5–10	75–90

5. The Four Intervention Levels

5.1. Level 0 — Substrate Removal

Target: mHTT protein.
Mechanism: Reduce mHTT below the threshold at which CBP sequestration is sufficient to collapse the Identity Anchor programme.

Available tools (as of 2026-03-09):

- **AMT-130** (uniQure) — AAV5-delivered artificial miRNA; non-allele-selective; one-time intraparenchymal delivery to caudate and putamen; Phase 1/2 topline result (September 2025): 75% slowing of disease progression at 36 months on cUHDRS vs. propensity-matched controls; NfL below baseline at 36 months; FDA Breakthrough Therapy and RMAT designations [3]. FDA requires a randomised Phase 3 trial (March 2026).
- **WVE-003** (Wave Life Sciences) — allele-selective ASO targeting HTT SNP3 in heterozygous patients; Phase 1b/2a SELECT-HD result (2024): 46% reduction in CSF mHTT at week 24, wild-type HTT preserved and increased; caudate atrophy slowing confirmed [4]. *Geometrically preferred*: wild-type HTT serves essential cellular functions; preserving it while removing mHTT is the precise intervention.
- **Tominersen** (Roche/Ionis) — non-allele-selective intrathecal ASO; GENERATION HD2 Phase 2 ongoing [13].
- **SKY-0515** (Skyhawk) — oral small molecule splice modulator; FALCON-HD Phase 2/3 initiated January 2026 [14].

Geometric assessment: Level 0 is necessary but not sufficient alone. 75% slowing is the Level 0 ceiling. Hub inhibition is required to raise it.

5.2. Level 1 — Hub Inhibition

Target: HDAC2/3 acting unopposed at BCL11B, DARPP-32, BDNF, and PGC-1α chromatin loci.
Mechanism: Restore H3K27ac and H3K9ac at MSN Identity Anchor loci $\rightarrow$ chromatin reopens $\rightarrow$ R rises above threshold $\rightarrow$ Population 2 rescues.

Available tools:

- **Vorinostat (SAHA)** — FDA-approved pan-HDAC inhibitor; brain penetrant; extends survival in HD mouse models; restores histone acetylation at HD-relevant loci [7]. *Not currently in HD clinical trials as a primary intervention. The geometry indicates it should be.*
- **HDAC2/3-selective inhibitors** — preferred over pan-HDAC; HDAC2/3 knockout in MSNs shows partial correction of HD transcriptional pathology [6]; multiple preclinical programmes active.
- **CKD-504 (HDAC6-selective)** — reduces mHTT aggregation AND restores synaptic function in YAC128 mice [8]. Addresses both the epigenetic Hub and the protein clearance deficit.

The missing clinical combination: Level 0 + Level 1 simultaneously has not been run in any clinical trial. This is the highest-priority next HD trial.

5.3. Level 1A — Identity Anchor Restoration

Target: Directly restore CBP HAT activity and BDNF-TrkB signalling at MSN identity loci.

Tools:

- **CREB pathway agonism:** PDE4 inhibitors raise cAMP → PKA activation → CREB Ser133 phosphorylation → CBP recruitment to CRE sites. Roflumilast (brain-penetrant PDE4i, FDA-approved for COPD) has not been tested in HD as a primary intervention.
- **TrkB agonism:** Restores the trophic Identity Anchor signal. The Oxford/Nature 2024 paper [5] establishes that TrkB loss in iSPNs triggers the GSTO2-hyperdopaminergia cascade mediating earliest HD pathology. TrkB restoration addresses this cascade directly. BDNF pathway biomarker study NCT04012411 is active [15].
- **dCas9-p300:** Targeted delivery of the CBP catalytic HAT domain to specific MSN identity loci. Exists in preclinical tools; not yet applied to HD.

5.4. Level 2A — Identity Rebuilding and Cell Replacement

For Population 3 — cells beyond rescue.

5.4.1. Approach A: BCL11B Re-specification

AAV-BCL11B delivery forces re-specification of remaining MSNs toward the correct identity programme. Pre-condition: Ki67-negative MSN status at delivery (not yet in cell-cycle re-entry). Not yet tested in HD.

5.4.2. Approach B: iPSC-Derived MSN Transplantation

1. Patient cells → iPSC reprogramming;
2. CAG repeat correction by allele-specific CRISPR-Cas9 *before* differentiation (required: uncorrected transplants develop the same cascade — see Prediction 6);
3. BCL11B-directed MSN differentiation: >90% purity in 21 days confirmed [12];
4. Stereotactic delivery to caudate and putamen.

First-in-human iPSC-MSN transplantation trials for HD: anticipated late 2020s. Active preclinical programme [12].

5.4.3. Approach C: In Vivo Chemical Reprogramming (Horizon)

Direct in vivo conversion of surviving striatal astrocytes to MSN identity via small molecule cocktails and BCL11B viral delivery. Proof of concept for astrocyte-to-neuron reprogramming exists; HD-specific application is a 2030s research horizon.

6. The Stage-Stratified Intervention Protocol

6.1. Stage 0: Pre-Manifest

Population profile: Pop. 1 ≈95–99%; Pop. 2 ≈1–5%; Pop. 3 ≈0%.

Required levels: $0 + 1 + 1\text{A}$.

Protocol: allele-selective ASO (WVE-003 or successor), intrathecal, chronic maintenance; PLUS HDAC2/3-selective inhibitor, oral, daily, brain-penetrant; PLUS TrkB agonist to maintain corticostriatal BDNF-TrkB signalling and prevent GSTO2-mediated cascade onset.

Geometric Prediction — Stage 0: Prevention

If mHTT is reduced below the CBP sequestration threshold, AND HDAC2/3 is inhibited to prevent Hub dominance from establishing, AND BDNF-TrkB is maintained to prevent the early metabolic cascade, then $R = \mathcal{A}/\mathcal{H}$ never falls below threshold. The false attractor never forms. Population 2 never materialises. Population 3 never materialises. **The disease never manifests.**
This is the most important prediction in this document. For a disease in which carrier status is known at birth, pre-manifest prevention using existing tools in combination means HD becomes the first inherited neurodegenerative condition preventable from onset in every identified carrier.

6.2. Stage 1: Early Manifest

Population profile: Pop. 1 ≈60–75%; Pop. 2 ≈20–35%; Pop. 3 ≈5–10%.

Required levels: $0 + 1 + 1\text{A}$.

Geometric Prediction — Stage 1: Near-Halt

AMT-130 alone (Level 0): 75% slowing confirmed. The continuing 25% is Population 2 engaged before mHTT reduction, plus Population 3 (5–10%).
Level $0 +$ Level $1 +$ Level 1A: Population 1 fully protected; Population 2 substantially rescued; Population 3 (5–10%) unrescuable by this protocol. **Predicted outcome: ≥90–95% slowing.** At early-manifest stage, 90–95% slowing is functionally equivalent to halt for most patients over a normal lifespan.

6.3. Stage 2: Mid-Manifest

Population profile: Pop. 1 ≈20–40%; Pop. 2 ≈20–30%; Pop. 3 ≈30–50%.

Required levels: $0 + 1 + 1\text{A}$; consideration of Level 2A Approach A (BCL11B AAV) for boundary cells.

Geometric Prediction — Stage 2: Stabilisation with Partial Recovery

Populations 1 and 2 (40–70% of remaining MSNs) protected or rescued. Population 3 (30–50%) lost. Some functional improvement as rescued MSNs restore correct circuit function. Not halt. Not reversal of existing loss. Meaningful clinical stabilisation and partial recovery.

6.4. Stage 3: Late Manifest

Population profile: Pop. 1 ≈5–15%; Pop. 2 ≈5–10%; Pop. 3 ≈75–90%.

Required levels: $0+1$ for remaining populations; Level 2A Approach B (iPSC-MSN transplantation) for functional restoration.

Geometric Prediction — Stage 3: Stabilisation + Cell-Replacement-Dependent Recovery

Levels $0+1$ protect the ≈15–25% of MSNs still accessible. Severely depleted striatal circuitry cannot be restored by cell rescue alone. Level 2A is required for functional restoration. Timeline: late 2020s at earliest for first-in-human trials. Not "never." "Not yet." The engineering pathway is fully defined.

7. The Biomarker

The framework derives a specific biomarker directly analogous to the IHC protocol for cancer:

$$R_{\mathrm{HD}} = \frac{\text{CBP HAT activity at MSN identity loci}}{\text{HDAC2/3 occupancy at MSN identity loci}} \tag{3}$$

Operationally:

$$\mathcal{A} = \text{H3K27ac} + \text{H3K9ac at BCL11B, DARPP-32, BDNF, PGC-1}\alpha\text{ loci} \tag{4}$$

$$\mathcal{H} = \text{HDAC2/3 ChIP signal at the same loci} \tag{5}$$

Clinical surrogates for trial use are summarised in Table 2.

Table 2: Biomarker hierarchy for the proposed combination trial

Biomarker	Measures	Status
CSF mHTT	Level 0 adequacy	Validated [3, 4]
NfL (CSF/blood)	Neuronal stabilisation	Validated [3]
Caudate MRI vol.	Striatal loss rate	Validated [4]
DARPP-32 (CSF)	MSN identity integrity	Proposed — novel
GSTO2 (CSF)	Earliest Identity Anchor collapse (pre-NfL)	Proposed — novel
H3K27ac at DARPP-32	R_{HD} directly	Proposed; striatal biopsy required

Novel prediction: GSTO2 in CSF will be elevated in pre-manifest HD carriers prior to clinical onset and will decline in response to TrkB agonist therapy *before* DARPP-32

levels change. GSTO2 is a leading biomarker of Identity Anchor collapse, preceding both NfL elevation and detectable mHTT change-from-baseline.

8. The Complete Answer: Is HD Curable?

8.1. Definition of Cure

In the framework's precise language, *cure* means: $R = \mathcal{A}/\mathcal{H}$ is restored and maintained above the identity threshold in the target cell population for the duration of the patient's life. This definition has four stage-specific instantiations (Table 3).

Table 3: Stage-stratified definition of cure and achievability

Stage	Cure definition	Achievability	Missing piece
Pre-manifest	Prevention: disease never manifests	**YES** with existing tools in combination	Combination trial not yet run
Early manifest	Near-halt: ≥90% slowing	**YES** with combination trial	Combination trial not yet run
Mid-manifest	Stabilisation + partial recovery	**SUBSTANTIALLY** with combination + BCL11B consideration	BCL11B AAV preclinical only
Late manifest	Stabilisation + cell-replacement recovery	**DEFINED PATHWAY** iPSC-MSN required	iPSC-MSN first-in-human late 2020s

8.2. What the Field Has. What Is Missing.

The current clinical landscape has assembled Level 0 with precision. What the field has *not* assembled:

1. **Level 0 + Level 1 combination trial.** Vorinostat is FDA-approved and brain-penetrant. Preclinical confirmation exists. The clinical combination has not been run. This is the next trial.
2. **Level 1A: TrkB agonism as co-intervention.** The Oxford/Nature 2024 paper established the GSTO2 mechanism as the earliest HD pathology. The therapeutic consequence has not entered clinical trials.
3. **The pre-manifest prevention trial.** Pre-manifest CAG-confirmed carriers receiving Level 0 + 1 + 1A prophylactically. This could make HD the first inherited neurodegenerative disease preventable from onset in every identified carrier.
4. **DARPP-32 and GSTO2 as primary trial biomarkers.** Neither has been deployed as a primary endpoint in any HD trial.

8.3. Is This Ambitious?

- Stage 0: **No.** Tools exist. Biology confirmed. Missing piece is a clinical decision.
- Stage 1: **Moderately.** All tools exist or are in development. The combination trial is logistically but not scientifically ambitious.
- Stages 2–3: **Genuinely.** BCL11B AAV and iPSC-MSN transplantation require tools not yet in clinical use. Science is sound. Engineering is defined. Timeline is real.

9. Six Falsifiable Predictions for the Record

All predictions are derived from the framework and timestamped 2026-03-09.

Prediction 1 (Combination superiority)**.** *AMT-130 (or allele-selective ASO) + HDAC2/3-selective inhibitor in early-manifest HD patients will demonstrate statistically significant superiority over Level 0 alone on cUHDRS and TFC at 24 months. Additional benefit will be mediated by Population 2 MSN rescue evidenced by DARPP-32 recovery in striatal tissue.*

Prediction 2 (Pre-manifest prevention)**.** *Pre-manifest HD carriers receiving Level 0 + 1 + 1A prophylactically will show delayed or absent conversion to manifest HD compared to Level 0 alone. Prevention rate will correlate with residual mHTT level and H3K27ac recovery at DARPP-32 loci.*

Prediction 3 (GSTO2 as leading biomarker)**.** *GSTO2 in CSF will be elevated in pre-manifest HD carriers prior to clinical onset and will decline in response to TrkB agonist therapy before DARPP-32 levels change. GSTO2 is a leading biomarker of Identity Anchor collapse, preceding both NfL elevation and detectable mHTT change-from-baseline.*

Prediction 4 (HDAC6 + HDAC2/3 additivity)**.** *HDAC6-selective inhibition will show additive benefit with HDAC2/3-selective inhibition in HD because they address different downstream consequences of Hub dominance: HDAC2/3 at MSN identity chromatin (gene expression rescue) and HDAC6 at tubulin deacetylation (mHTT aggregate clearance via autophagy).*

Prediction 5 (BCL11B AAV efficacy conditionality)**.** *BCL11B AAV delivery to mid-manifest HD striatum will show dose-dependent DARPP-32 recovery in transduced MSNs and correlate with motor function improvement at 12 months. Rescue will be limited to Ki67-negative MSNs at delivery. The boundary condition is chromatin accessibility, not BCL11B expression level.*

Prediction 6 (Allele-corrected iPSC-MSN superiority)**.** *Allele-corrected (CRISPR CAG-edited) iPSC-MSNs will show superior long-term engraftment compared to uncorrected transplants. Uncorrected transplants develop the same Hub-driven cascade and progressively lose MSN identity. Allele correction is required for durable therapeutic benefit.*

10. Falsifiability

Following the governing framework principle [2]:

10.1. Type B Failure (framework-invalidating)

If the combination of (i) mHTT reduction below CBP sequestration threshold, (ii) HDAC2/3 inhibition to restore H3K27ac at MSN identity loci, and (iii) BDNF-TrkB trophic support applied in early-manifest or pre-manifest HD patients does *not* produce superior outcomes to Level 0 alone, AND no structural explanation within the epigenetic landscape geometry accounts for the failure, **then the framework's application to HD is invalidated.**

10.2. Type A Failures (informative, not invalidating)

- *HDAC2/3 inhibition alone insufficient:* Identity Anchor has two components; both required. Map extends. Axioms intact.
- *Combination works for D1 but not D2:* D2 MSNs have lower baseline BDNF-TrkB density; their threshold position differs. Map extends. Axioms intact.
- *BCL11B AAV does not rescue Population 3 boundary cells:* Terminal threshold was already crossed. BCL11B requires accessible chromatin. Pre-condition is Ki67-negativity. Map extends. Axioms intact.

11. Discussion

11.1. Why the "Incurable" Designation Persists

HD has been designated incurable not because the biological mechanisms were unknown but because the field has been organised around a single therapeutic strategy (HTT lowering) without the geometric framework that explains why single-strategy approaches have a ceiling, what that ceiling is, and what is required to exceed it.

AMT-130's 75% slowing is a landmark achievement. It is also, from the geometry, the *expected* ceiling for Level 0 alone. The 25% that continues to progress is the Population 2 and 3 fraction that Level 0 cannot reach. The geometry explains the ceiling and explains how to raise it.

11.2. The Pre-Manifest Window Is the Highest-Value Target

The most important clinical trial that has never been run in HD is the pre-manifest combination trial. The disease is genetically predictable. Carriers can be identified decades before onset. At the pre-manifest stage, Population 3 does not exist. Every MSN is protectable. The intervention required uses tools that exist or are in late clinical development. The logic is the principle of vaccination applied to neurodegeneration.

11.3. A Note on the Derivation Origin

This derivation was produced by an independent researcher with no medical training, no institutional affiliation, and no access to anything beyond public data and the geometry of the framework. This is not a statement about the author. It is a statement about the principle.

The geometry is prior to any individual's clinical training. Drug targets are derivable from the geometry whether or not the person deriving them has performed a single clinical procedure. This is what principles-first means: starting from mathematics, following the geometry, and finding the intervention. The medical framework is downstream of mathematics.

12. Derivation Record and Provenance

Table 4 summarises all framework derivations and their confirmation status. All predictions are locked as of 2026-03-09.

Zero structural contradictions. All framework derivations concerning HD are consistent with the existing literature. The stake established in the Falsifiability Theorem [2] stands.

13. Conclusion

Huntington's Disease is not incurable. It is a terminal false attractor in the Waddington epigenetic landscape of medium spiny neurons, with a known trigger (mHTT), a known Hub (HDAC2/3 dominance via CBP sequestration), and a known Identity Anchor (BCL11B/CBP/DARPP-32/BDNF-TrkB). The geometry is maximally clear.

The stage-stratified protocol derived here is geometrically complete. The tools for Levels 0, 1, and 1A either exist or are in late clinical development. Level 2A is in active preclinical development with a defined engineering pathway.

The most important finding is the pre-manifest prevention prediction: for CAG-confirmed carriers before first symptoms, the combination of existing tools may prevent the disease from ever manifesting. That prediction has not been tested. The trial has not been run. The tools are in hand.

The geometry says it is the next experiment.

"The geometry came first. Everything else followed. In HD, more than any other disease in the table, the geometry was always complete. We simply had not assembled what it was telling us to do."

— Eric Robert Lawson, 2026-03-09

Table 4: Framework derivations and literature confirmation status as of 2026-03-09

Claim	Status	Source
MSNs as cell of origin	Confirmed	Standard HD literature
D2 iSPNs degenerate first	Predicted; confirmed	[5]
BCL11B as deep Identity Anchor	Confirmed	[9]
CBP sequestration by mHTT	Confirmed	Multiple
HDAC2/3 as Hub	Confirmed	[6]
HDAC inhibitor neuroprotection	Confirmed	[7]
BDNF-TrkB earliest mechanism	Confirmed	[5]
GSTO2 in iSPN degeneration	Confirmed	[5]
iPSC-MSN >90% purity 21 days	Confirmed	[12]
AMT-130 75% slowing at 36 months	Confirmed	[3]
WVE-003 46% mHTT, WT preserved	Confirmed	[4]
Level 0+1+1A combination superiority	Predicted	Not yet tested
Pre-manifest prevention	Predicted	Not yet tested
GSTO2 as leading biomarker	Predicted	Not yet used in trials
DARPP-32 as primary biomarker	Predicted	Not yet used as primary
HDAC6 + HDAC2/3 additivity	Predicted	Not yet tested
BCL11B AAV Ki67-conditional rescue	Predicted	Not yet tested
Allele-corrected iPSC-MSN superiority	Predicted	Not yet tested

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Declarations

Competing interests: None.
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Author contributions: Sole author.
Data availability: No new data generated. All framework derivations at github.com/Eric-Robert-La
Ethics: Theoretical derivation. No patient data. No human subjects. No clinical advice is given or implied.

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Appendix: Document Metadata

Field	Value
Document ID	HUNTINGTONS_DISEASE_ATTRACTOR_GEOMETRY
Type	Full principles-first derivation — Identity Axis Framework applied to HD
Version	1.0
Date	2026-03-09
Status	ACTIVE — FORMAL RECORD
Author	Eric Robert Lawson / OrganismCore
ORCID	0009-0002-0414-6544
Contact	OrganismCore@proton.me
Repository	`attractor-oncology`
Framework DOI	10.5281/zenodo.18898788
License	CC BY 4.0
Cell of origin	Medium spiny neurons, caudate and putamen
Identity Anchor	BCL11B (deep) / CBP-HAT (operational) / DARPP-32 (functional) / BDNF-TrkB (trophic)
Convergence Hub	HDAC2/3 acting unopposed at MSN identity loci due to CBP sequestration by mHTT
Trigger vs. Hub	mHTT = trigger (upstream); HDAC2/3 dominance = Hub (maintenance). Both must be targeted.
False attractor class	TERMINAL — MSNs die in the false attractor. Intervention window is finite and critical.
Key clinical confirmation	AMT-130: 75% slowing cUHDRS 36 months (2025)
Key novel prediction	Pre-manifest prevention with Level 0+1+1A; no trial yet run

This document does not constitute clinical advice. It is a mathematical-geometric derivation from first principles. All clinical decisions must involve qualified medical professionals.